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Recent progress in dynein structure and mechanism

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Dynein is a minus-end-directed microtubule motor crucial to diverse cellular processes. The unwieldy size of the molecule and the difficulty of expressing and purifying mutants have hampered mechanistic studies of dynein. Recent progress sheds light on key unsolved questions concerning how the molecule is really organized, what conformational changes accompany ATP hydrolysis and whether two or three motor domains are coordinated in their motion.

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Introduction

Forty years have passed since dynein was first purified from *Tetrahymena* cilia as a force-generating ATPase [1]. Dynein ATPases are now known to form a morphologically uniform but functionally diverse protein family whose members are crucial to diverse cellular processes. Dyneins contain one to three heavy chains: each heavy chain consists of a C-terminal globular head together with two elongated flexible structures called the stalk (the microtubule-binding domain) and the N-terminal tail (the cargo-binding domain) (Figure 1). Dyneins are categorized into two major classes, axonemal and cytoplasmic.

The axonemal dynein produces the bending motions that propagate along cilia and flagella (reviewed in [2,3]). They are localized in the outer doublet microtubules in the axoneme and form projections called dynein arms, each of which is composed of multiple dynein molecules. These dynein molecules are organized such that a few heavy chains form heterodimers, heterotrimers or monomers. Heavy chains have different amino acid sequences, each with distinct functions; all of these functions are crucial for proper axonemal operation [3,4] and even enable the generation of torque [5,6] and oscillation [7].

Cytoplasmic dynein drives a variety of fundamental cellular processes, including nuclear migration, organization of the mitotic spindle, chromosome separation during mitosis, and the positioning and function of many intracellular organelles (reviewed in [8–10]). Cytoplasmic dynein is thought to be a homodimer with two heavy chains, each of which comprises a head domain joined to a tail. This cytoplasmic dynein contains several accessory subunits, termed intermediate, light intermediate and light chains.

The aim of this review is to summarize recently revealed structural and mechanical properties of these dyneins.

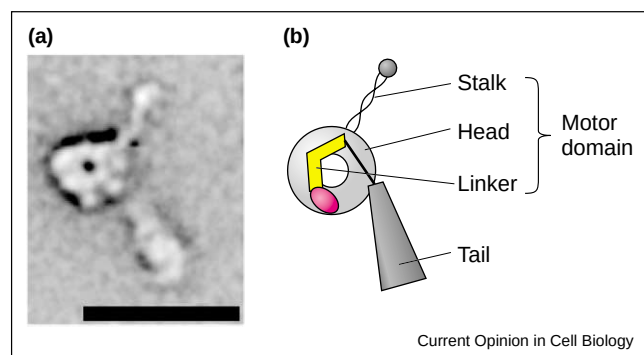
The structure of a motor domain

The absence of any crystal structure makes dynein a poor cousin to kinesin and myosin. On the basis of biochemical and molecular biological studies, it has been shown that the dynein heavy chain (molecular mass >500 kDa) contains a large fundamental motor domain in the C-terminal 380-kDa fragment [11,12,13^{*}], incorporating sites of ATP hydrolysis and microtubule binding. Sequence analysis shows that dynein is a member of the AAA⁺ protein superfamily (where AAA stands for ATPase associated with various cellular activities) [14]; its similarity to the other members of this family lies in the characteristics of its ATPase domain secondary structure as well as in its primary structure (reviewed in [15]). The electron microscopic observations [16,17^{*},18^{**},19^{*}] show that dynein has a complex morphology, in which six sequentially linked AAA⁺ ATPase-like domains (AAA modules), each of 35–40 kDa, are arranged in a ring, probably together with a seventh domain.

The first four AAA modules each contain a highly conserved Walker A motif (GXXGXXGKT/S, also called a P-loop) and a Walker B motif (DEXX) [20–23], and each module is thought to bind a nucleotide [24,25]. A principal site of ATP hydrolysis had been previously mapped to the first Walker motif by vanadate-mediated photocleavage of the heavy chain of axonemal dynein [26]. Further strong support for a functional role for the first Walker motif is now provided by molecular dissection of cytoplasmic dyneins in which mutation of the P-loop abolishes their motor activities [27^{**},28^{*}].

It seems likely that the additional Walker motifs act in a regulatory manner by binding either ADP or ATP. In some dynein isoforms the presence of ADP is known to be essential for motile activity *in vitro*, and in others ADP increases the velocity of microtubule movement driven by dynein [29,30], indicating that ADP binds to at least

Figure 1



Structural organization of single-headed axonemal dynein.

(a) Dynein-c purified from *Chlamydomonas* flagellar axoneme in the apo state was observed by negative-stain electron microscopy followed by single-particle image processing (reproduced with permission from [19]). The head has seven lobes of density forming an asymmetric ring. Two elongated structures extend from the head ring. The stalk carries at its distal end the microtubule binding domain. The tail (called the stem in earlier papers) is the cargo binding domain. Scale bar: 30 nm. **(b)** A schematic drawing of the dynein molecule indicates that motor domain is composed of the stalk, the head ring and the putative linker (colored yellow).

one of these AAA modules. A hypothetical atomic structure produced by homology modeling of the dynein AAA modules suggests that the nucleotide-binding Walker A motifs lie close to the interface between adjacent modules [31]. Interactions between adjacent AAA modules via their nucleotide pockets supports the idea that they may act in concert to produce a functional motor [32].

The function of these additional Walker motifs has now been elucidated further by molecular dissection in expressed cytoplasmic dynein [27[•],28[•]] and by ATPase measurements of truncated cytoplasmic dyneins and their mutants expressed in *E. coli* [32[•]]. Manipulating the Walker A motif of the third AAA module in cytoplasmic dynein showed that the resulting construct failed to be released from tight-binding to microtubules by ATP [27[•],28[•]] and lost most of its microtubule-activated ATPase activity [27[•]]. The motor activity of the construct was retained although the velocity of microtubule sliding was less than a twentieth of the velocity associated with the wild type [27[•]]. Mutations in the second and fourth Walker motifs did not affect the apparent binding affinity for microtubules in the presence of ATP, but reduced both ATPase and motor activity [27[•]]. These results clearly indicate that the additional Walker motifs regulate the motor activities of dynein through nucleotide binding and hydrolysis.

The stalk

The stalk is up to 15 nm in length and is an anti-parallel coiled-coil that lies between the fourth and fifth AAA

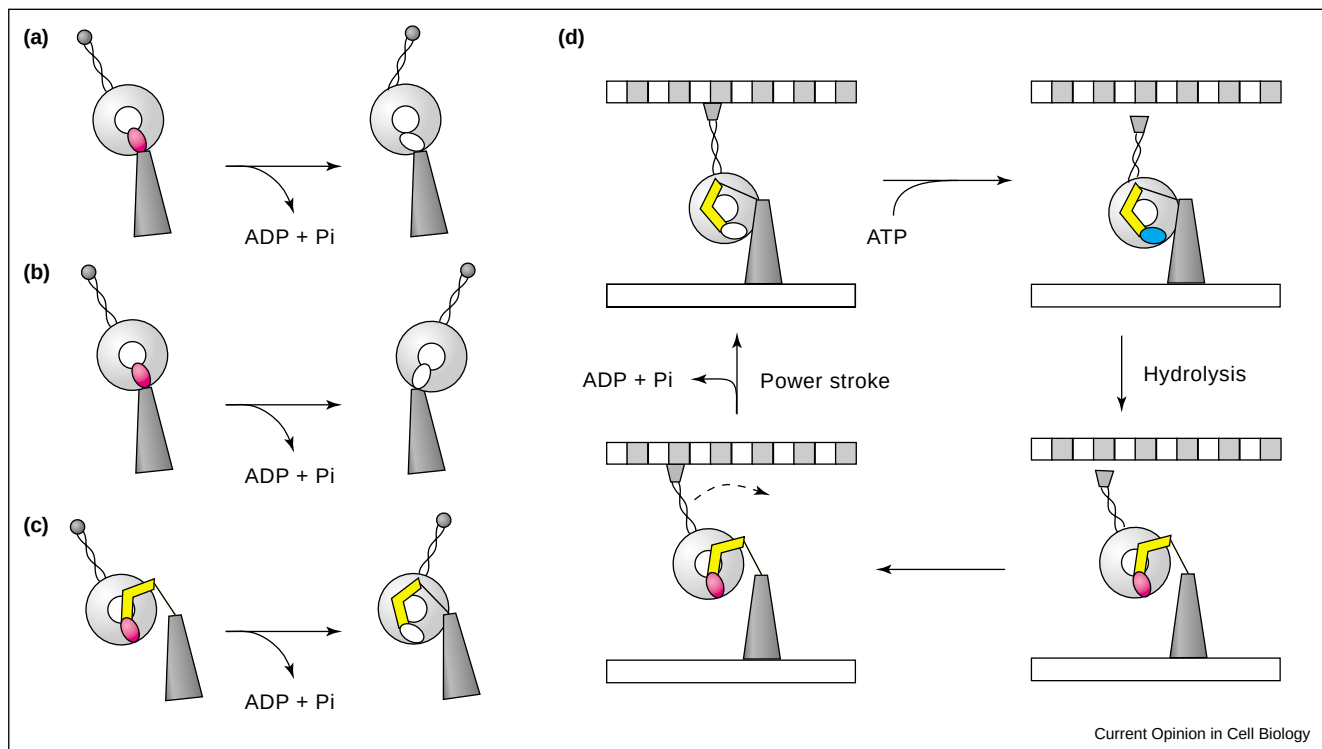
modules [11,33]. A small globular domain at its tip is the ATP-sensitive microtubule-binding domain [11,12], in which there is poor sequence conservation [34]. The microtubule-binding region of the stalk tip was recently examined with a recombinant stalk tip peptide [35^{••}]. The stalk tips were observed binding to tubulin dimers with a periodicity of 8 nm and sharing the binding region with kinesin. Since the microtubule-binding site of dynein is segregated away from the primary site of ATP hydrolysis, the architecture of dynein is remarkably different from that of the well-characterized motor proteins, kinesin and myosin.

Although the structure of the stalk is not fully understood, it is predicted to transmit conformational changes along its length, implying a requirement for dynamic changes to the helix-helix interactions. More specifically, binding of ATP to the first AAA domain causes the dissociation of dynein from the microtubule. We recently found that stalk flexibility was reduced in apo-molecules, indicating that this domain has greater stiffness [18^{••},19[•]]. Thus, some mechanism must exist that triggers changes in the stability of the coiled-coil, and hence in the stiffness of the stalk.

Sequences of left-handed coiled-coils are characterized by a seven-residue periodicity (heptad repeat) represented by the expression (a-b-c-d-e-f-g), and sequences of the stalk region have this periodicity [36]. Apolar residues preferentially appear in the first (a) and fourth (d) positions of the heptad. A 'knobs-into-holes' packing of the apolar side chains into a hydrophobic core is the main mechanism by which coiled-coils are stabilized. On the basis of the amino acid sequences, a discontinuity in the heptad repeat, of a half-pitch shift, is predicted [37] (see Figure 2b in [34] halfway along the outward α -helix of the stalk of cytoplasmic dyneins. In contrast, the heptad repeat is highly ordered throughout the inward α -helix. This discontinuity probably modulates the stability of the coiled-coil of the stalk, although large-scale melting of a coiled-coil in the nucleotide-bound state seems unlikely. In nucleotide-bound states electron microscopy of axonemes suggested that the stalk remains intact and maintains contact with the microtubule even when the axonemes are fixed in the presence of ATP [38,39], and the stalk stays the same length in isolated dynein [18^{••}].

A coiled-coil peptide has a flexural rigidity of the order of 400 pN·nm² [40]. It will deflect under a force F by a distance y like a rigid beam of length L where $y = FL^3/3EI$, with E being the Young's modulus and I the second moment of inertia of the cross-section. The product of E and I is the flexural rigidity. Thus, a 15nm length of coiled-coil peptide clamped at one end has a stiffness of ~ 0.36 pN·nm⁻¹. Since the range of conformations for an axonemal dynein molecule observed in negatively

Figure 2



Proposed mechanisms of power stroke of dynein. Release of Pi and ADP from the catalytic site causes (a) tilting of the stalk [17^{*},31,44] or (b) tilting of the tail [34]. (c) Alternatively, the first AAA domain swings a putative linker connecting the first AAA domain and the tail [18^{**}]. The small conformational change between the first AAA domain and its neighbors is amplified by the linker and possibly stabilized by docking onto the head. (d) A speculative model for the mechano-chemical cycle of dynein based upon the EM observations [18^{**},43]. The ATPase mechanism was most thoroughly studied with three-headed 22 S dynein of *Tetrahymena* axonemes (reviewed in [62]). We start at the state where dynein without nucleotide tightly binds to a microtubule. ATP binds to the motor domain and this binding promotes dissociation of dynein from the microtubule. The detached dynein quickly hydrolyzes ATP and re-primers its conformation for a power stroke. Microtubule movement is then initiated by tight binding to the tip of the stalk, which promotes a concerted conformational change in the head ring, thereby activating release of ADP and phosphate from the ATP binding site. Force generation and the power stroke follow microtubule binding (from the lower left to the upper left panel). Thus, rigid coupling among several mechanical components causes a rolling of the head towards the tail. The first AAA domain is colored white in the no-nucleotide state, blue in the ATP state and red in the ADP-Pi state.

stained samples [18^{**},19^{*}] depends on the distribution of thermal energy, this range enables estimation of stiffness of the stalk. The stiffness was estimated to be $0.5 \text{ pN}\cdot\text{nm}^{-1}$ in apo molecules and $0.14 \text{ pN}\cdot\text{nm}^{-1}$ in ATP-vanadate molecules [41]. A transverse force of 1–5 pN corresponding to the maximal force of dynein [7,42,43] applied to the free end of the coiled-coil deflects it through a distance of $\sim 3\text{--}14 \text{ nm}$. The negative-stained electron microscope images also indicated that the tail of the dynein molecule is flexible with a stiffness of $0.1 \text{ pN}\cdot\text{nm}^{-1}$ [41]. Since the stiffness of the stalk and tail are in series, this gives a total stiffness of $0.05\text{--}0.08 \text{ pN}\cdot\text{nm}^{-1}$ [41], which is similar to that estimated by optical trap experiments [43]. Thus dynein is a very flexible molecule, suggesting that under low- or zero-load conditions it may be able to take steps much larger than the mean step size of 15 nm seen by electron microscopy [19^{*}].

Conformational changes of dynein coupled with ATP hydrolysis

Judging from the conserved sequences and structural similarity between axonemal dynein and cytoplasmic dynein, it is reasonable to suppose that both dynein subfamilies adopt the same mechanism for force generation. At least three models have been proposed for force generation of dynein (Figure 2). First, rotation about the junction between the head ring and the stalk resulting from the ATPase-coupled rearrangement of the AAA modules in the head ring is proposed to swing the microtubule-binding stalk relative to the tail (Figure 2a) [17,31,44]. The second model suggests that ATPase-coupled rotation about the junction between the head ring and the tail again swings the head and the microtubule-binding stalk (Figure 2b) [34]. In the third model, coordinated multiple conformational changes amplify small conformational changes between the first AAA

domain and its neighbors by docking the head ring onto a linker which connects the tail and the head ring [18^{••}]. The resulting rotation of the head ring swings the stalk (Figure 2c).

In spite of a Gaussian distribution of the stalk angle of axonemal dynein [18^{••},19[•]], the stalk of cytoplasmic dynein showed three discrete angular distributions in the apo state [17[•]]. The angular changes in stalk position appear to be correlated with positional shifts among the seven main ring densities. The probabilities of observations in three angular states were, however, almost equal, suggesting (according to Boltzmann's law) that energy differences among three states are 0.2–1.3 $k_B T$. If these angular changes reflect different mechanochemical states of dynein, these energy differences seem too small to assign them to conformational changes coupled with ATP hydrolysis.

The key structure of the third model is the linker [18^{••}]. If this mechanism is common to all dyneins, the linker domain sequences are likely to be conserved among them. Vanadate photocleavage of recombinant dynein heavy chains [11,12] showed that essential sequences are contained in residues 1137–1455 in rat cytoplasmic dynein. A conserved element, R/KXRHWXXI/L, is found within the essential sequence. Although the potential contribution of this sequence to either dynein structure or function is not clear, it might make up part of the linker.

Mechanical properties of dyneins

With an optical trap technique, the mechanical properties of several dyneins have been measured with spatiotemporal resolutions of nanometers and milliseconds. The single-headed inner-arm dynein-c showed processivity and a step size of 8 nm [43]. The outer-arm dynein of *Tetrahymena* cilia — a heterotrimeric 22S dynein — showed processive movement with a step size of 8 nm at low concentrations of ATP, while at higher concentrations it showed pulse-like force generation similar to what is observed in skeletal muscle myosin [42]. In both cases, these dyneins often missed a forward step and showed backward slippage under high loads [42,43]. Recently, the movement of a single cytoplasmic dynein was measured and its step size decreased from 32 nm to 8 nm with increasing load or ATP concentration [45[•]].

Some inner-arm dyneins have the ability to generate torque that causes microtubules to rotate during forward translocation [5,6]. Cytoplasmic dynein-coated beads exhibited larger and more frequent lateral movements between microtubule protofilaments than was observed with kinesin [46]. Hence, they avoid being confined to one protofilament.

These mechanical experiments raise many questions. For instance, how does the monomer move processively?

Processivity of a single-headed motor on microtubules has previously been reported for the kinesin-like motor KIF1A [47,48]. In this case, processivity is achieved through the electrostatic interactions between the negatively charged so-called 'E-hook' of tubulin and the positively charged K-loop of KIF1A [49]. A similar mechanism may apply to dynein-c. Indeed, an outer arm dynein was reported to interact with microtubules in a way that permits the latter to diffuse only along their longitudinal axes [50]. However, the equivalent of the K-loop sequence has not been detected in dynein.

Additional questions include how the dimer or the trimer coordinates its multiple motor domains during stepping, and how step size changes with load. In studies of other processive motors, such as myosin V (reviewed in [51]) and conventional kinesin (reviewed in [52]), a mechanism has been proposed in which two identical motor domains alternatively swing forward at each step while one motor domain remains attached on the track. The motor domains alternatively perform the identical tasks. The experimental support for this hand-over-hand mechanism has accumulated [51].

However, dynein differs significantly from kinesin or myosin with its larger size, its apparent freedom from a requirement to walk along the same track and its great flexibility. Although the physiological relevance of dynein's flexibility is not presently understood, given its large size and suggested power stroke in relation to the tubulin lattice [18^{••},19[•]] it could perform variable stride lengths even without an alternative swing of the large motor domains. This is in contrast to hand-over-hand stepping since the step size of a motor except myosin VI [53] can be predicted to be proportional to the length of its power stroke [54,55]. Furthermore, the motor domains of dynein heteromers do not need to behave in the same manner. Hence, the processive movement of dynein could be described simply by an inchworm model, in which dynein moves in inching gait by extending its leading head and bringing the rear to meet it, coupled with diffusion search.

In an axoneme, the activity of dynein molecules propagates through linear or two-dimensional arrays of dynein molecules closely packed on peripheral doublet microtubules. The behavior of axonemal dynein is integrated on a large scale, causing flagellum beating and wave propagation. Several important studies on this kind of concerted operation of multiple dyneins (including mixtures of different types of dyneins) have been reported. A solitary dynein arm on an extruded doublet microtubule was capable of oscillatory movement [7]. Axonemes immobilized on a glass surface along their entire length showed a back-and-forth oscillation of a bundle of outer doublets at high frequencies driven by dyneins [56]. This longitudinal oscillation of the axoneme was coupled with

transverse oscillation [57]. In addition, realistic computations of the coordinated motion of cilia and flagella were carried out based upon solid evidence of the mechanics and biochemistry of dynein and axonemes [58–61]. In these computations, the force exerted by dyneins on elastic components of the axoneme was evaluated and the motions of axonemes were computed. These computations will provide us new predictions that cannot be obtained by simple descriptions of mechanical parts and their relations.

Conclusions

Data presented to date indicate that the structure and force-generating mechanisms of dynein are quite different from those of the other linear motors, kinesins and myosins. We now have a powerful armory of techniques enabling the expression of recombinant cytoplasmic dynein with full motor activity [13•] and molecular dissection of cytoplasmic dynein [27•,28•]. Although axonemal dynein awaits the establishment of an expression system, this armory will bring us to a better understanding of the dynein conformational cycle during motility and of the mechanism for transmission of conformational changes through the stalk.

Important questions awaiting further work concern the possible coordination between two or three motor domains connected by a presumably flexible tail and the mechanical feedback or modulation of force generation in dynein, which may play an important role in the bending and propagation of flagella. To describe the coordinated sequence of conformational changes and kinetics of two heads, single-molecule studies combined with the expression of genetically engineered dyneins will be a powerful experimental approach.

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